

2023



WPS Office
Make It Simple

Actively Embracing AI Using Amazon SageMaker with Kingsoft Office | Case Study | AWS

Learn how Kingsoft Office in software-built machine learning models to predict rates for paying users using AWS.

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40%

Over 40% reduction in storage costs



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1 50%



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Overview

Established in 1988, Kingsoft Office, a software company, it has been deeply engaged in the research and development of office software. Adhering to the brand concept of "Inspiring Wisdom," the company is committed to bringing the simplest and most efficient office experience to individuals, families, and organizations. It helps individual users to create and live more simply and happily, and organizations operate more efficiently and safely. As of October 2023, its flagship product, WPS Office, has over 100 million monthly active users in the global market and has been adapted to over 46 languages, becoming one of the world's mainstream office software products.



Opportunity | Bringing New Opportunities for Office Software with Rapid Overseas Growth

Artificial intelligence (AI) has sparked a new wave of technology. The combination of AI and office software will greatly improve user experience, work efficiency, backend management operations, and many other aspects of office systems.

[WPS Office](#) (WPS), the subsidiary of Kingsoft Office, is a pioneer of office software. It not only entered the era of AI but also went global very early, actively expanding new business in overseas markets.



entered the international market in 2007, WPS Office has continued to provide stable and high-quality office services to users in over 220 countries and regions worldwide.

However, the rapid growth of business has brought both opportunities and challenges to Kingsoft Office. WPS hoped to continually update product forms and iterate product functions to better meet the needs of overseas users, and for global user access capability, WPS needed stable and reliable cloud infrastructure support. The company was also looking to reduce storage costs and increase user access efficiency with cloud products. In the process of business expansion and rapid user growth, Kingsoft Office has been thinking about the use of data and AI capabilities to drive service-level improvement, reshape the user experience, and enhance user retention. To meet these needs, Kingsoft Office turned to Amazon Web Services (AWS), and the resulted to reduced storage costs and new breakthroughs in AI capabilities.

“Since entering the overseas market, Kingsoft Office has used the comprehensive data, AI capabilities, and industry experience of AWS to create refined operations, reduce storage costs by over 40 percent, and provide assistance for business development,” says Zhang Ning, global vice president of Kingsoft Office.

Solution | Promoting Overall User Growth with Data and AI Capabilities

Since initiating its overseas operations, WPS has been in collaboration with AWS, marking a long-standing partnership. WPS, the WPS official global website, and the desktop version of WPS Application of Kingsoft Office all run stably on AWS.

As the flagship product of Kingsoft Office, WPS has extremely strict requirements for data storage and management. WPS stores petabytes of data, which brings enormous pressure on storage cost. Meanwhile, since WPS provides users with timely access to cold data, archived storage requires higher standards for data extraction time. According to product requirements, filtering out the data and converting fine-grained storage types are needed. However, this may cause high costs of research and development and maintenance.

With the assistance of [AWS Enterprise Support](#) and AWS Solutions Architecture team—which provides a comprehensive suite of resources—WPS has designed a full lifecycle management strategy, to reduce latency and storage costs for end users. This strategy is realized by comprehensive categorizing of user documents based on aspects such as access frequency, most recent access time, and modification time. Finally, an automated lifecycle management process was established through [Amazon Simple Storage Service](#) (Amazon S3), which is an object storage service offering cutting-edge scalability, data availability, security, and performance. Data in Amazon S3 that had not been accessed for over 365 days and had a file size greater than 128 KB was archived in [Amazon S3 Glacier Instant Retrieval](#), which is an archive storage class that delivers the lowest-cost storage for long-lived data that is rarely accessed and requires retrieval in milliseconds. In the end, nearly half of WPS’s storage entered Amazon S3 Glacier Instant Retrieval, improving data access speed and optimizing storage costs. (See Figure 1: Architecture Diagram of WPS Office Document Business on AWS below.) Amazon S3 realizes the intelligent layering of storage through automated lifecycle management, achieving over 40 percent cost optimization in storage without compromising the product experience.

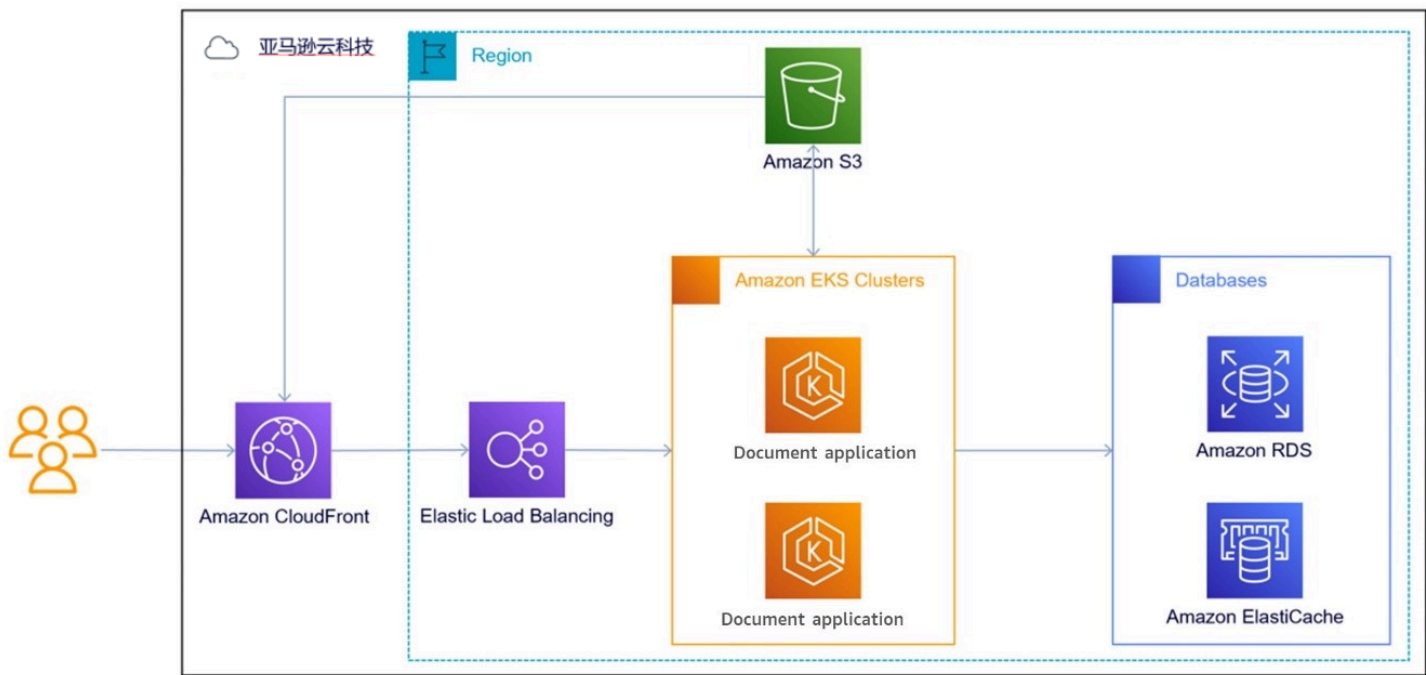


Figure 1. Architecture Diagram of WPS Office Document Business on Amazon Web Services

For business data storage and processing, Kingsoft Office uses [Amazon RDS for MySQL](#), which helps to set up, operate, and scale a relational database in the cloud with just a few clicks. With a simple API call, it can flexibly expand computing resources or storage capacity associated with database instances.

To accurately predict the purchasing tendency of overseas users, Kingsoft Office uses [Amazon SageMaker](#)—a fully managed service that provides every developer and data scientist with the ability to build, train, and deploy machine learning (ML) models quickly—during the product trial period to design and train a machine learning model by analyzing user registration, user activation of trial functions, payment type, and other behavioral data to predict the possibility of conversions.

The overall process of this prediction service is triggered by daily tasks through [Amazon EventBridge](#), which is used to build event-driven applications at scale. After the processing is completed, the data is synchronized to the operating system of Kingsoft Office through [AWS Lambda](#), which is a serverless, event-driven compute service. According to the predicted data, operators initiate corresponding operational strategies for users in different circles, thereby improving the conversion rate. (See Figure 2: Architecture Diagram of Kingsoft Office Purchase Prediction on AWS.)

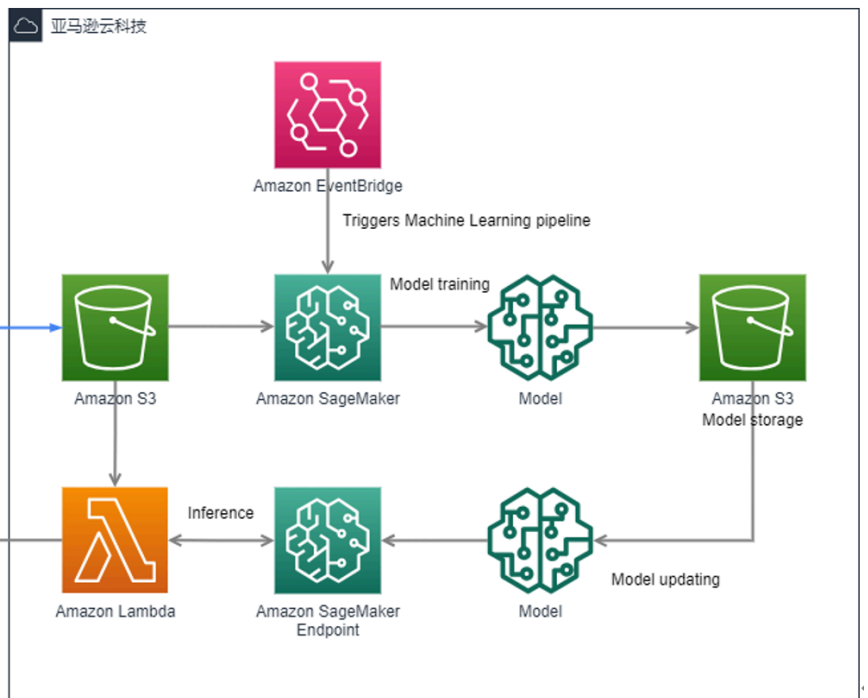


Figure 2. Architecture Diagram of Kingsoft Office's Overseas User Purchase Prediction on Amazon Web Services

With the rise of generative AI in 2023, Kingsoft Office is committed to integrating AI technology into office suites to improve user efficiency and enhance the user experience. In response to this demand, the AWS account team, in collaboration with the AWS GCR Data Lab and the AWS Rapid Prototyping team are providing suggestions for AI innovation and helping to complete the proof of concept tests of vector search and stable diffusion packaging solutions. Moreover, Kingsoft Office is actively coordinating with AWS global team, to be the first to trial its recently launched fully managed generative AI modeling tool, Amazon Bedrock.

"We have collaborated with AWS Generative Artificial Intelligence Innovation Center (GAIIIC) to conduct comprehensive testing of various office application scenarios using Amazon Bedrock," says Ning. "We are satisfied with the model performance of the product and look forward to mutually beneficial cooperation with AWS to bring positive results."

Outcome | Embracing the Power of AI to Facilitate Innovation

Kingsoft Office utilizes AWS Availability Zones in the United States, Japan, and India for its operations. This makes its products more accessible to users and enhances end-user experience. After using the Amazon CloudFront a global Content Delivery Network, the latency of Kingsoft Office in Mumbai, India, has decreased and stabilized from more than 1 second to less than 500 milliseconds.

For the data ingestion of WPS desktop applications, Kingsoft Office hopes to migrate the data ingestion workload to AWS. That could reduce system costs by utilizing resource elasticity, price and performance advantages of AWS.

"On Amazon SageMaker, we have built a complete set of user refinement operation paths from user identification to either user conversion or lost user recovery, and we will gradually extend to explore more user scenarios in the future," says Yao Huapeng, global technical director of Kingsoft Office.

In the foreseeable future, the products and functions of Kingsoft Office software will be enriched. Kingsoft Office will continue using AWS to innovate the office system. Kingsoft Office looks forward to using AWS services for document translation, optical character recognition (OCR), and intelligent customer service.

About Kingsoft Office

Established in 1988, Kingsoft Office is one of the world's leading suppliers of application software products and services. It has more than 570 million monthly active user devices around the world, including governments, enterprises, and individuals. Its flagship product, WPS Office, supports all mainstream systems, multi-device cross-screen, multiplayer online collaboration, and AIGC, providing services for individual and enterprise users.

AWS Services Used

Amazon SageMaker

Build, train, and deploy machine learning (ML) models for any use case with fully managed infrastructure, tools, and workflows.

[Learn more »](#)

Amazon S3 Glacier Instant Retrieval

Amazon S3 offers a range of storage classes that you can choose from based on the performance, data access, resiliency, and cost requirements of your workloads.

[Learn more »](#)

AWS Enterprise Support

AWS Enterprise Support provides a comprehensive suite of resources, including proactive planning, advisory services, automation tools, communication channels, and 24/7 expert support.

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Amazon RDS



Set up, operate, and scale a relational database in the cloud with just a few clicks.

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